

THE LAST MILE CHALLENGE

Uncovering Delivery Delays, Bottlenecks, and Operational Risks Through Data Analytics

Project Master Documentation

Version: 1.0

Project Type: Business Analytics Portfolio Project

Domain: Logistics & Supply Chain Analytics

Prepared By: Namratha Padiyar

Purpose of this Document

This document serves as the master technical and business documentation for **The Last Mile Challenge**. It has been prepared to preserve every significant aspect of the project, including its objectives, implementation decisions, analytical workflow, business reasoning, technical execution, and final outcomes.

Unlike an executive report that focuses only on findings, this document captures the complete project lifecycle—from understanding the business problem and preparing the dataset to performing SQL-based analysis, creating Python visualizations, and translating analytical results into actionable business recommendations.

The objective is to maintain a single source of truth that enables the project to be understood, reproduced, explained, and extended without referring back to the original working files. It is intended to support future portfolio updates, resume preparation, interview discussions, GitHub documentation, LinkedIn content, and knowledge retention.

Revision History

Version	Date	Description
1.0		Initial Release First complete documentation of The Last Mile Challenge project.

Table of Contents

1. Project Identity
2. Executive Summary
3. Business Background
4. Problem Statement
5. Dataset Documentation
6. Data Quality Assessment
7. Data Cleaning Strategy

8. Technical Implementation
9. SQL Business Analysis
10. Python Visualization Workflow
11. Business Insights
12. Strategic Recommendations
13. KPI Documentation
14. Challenges Faced
15. Lessons Learned
16. STAR Interview Version
17. Resume Master Description
18. ATS Keywords
19. Skills Demonstrated
20. Future Enhancements
21. Portfolio Assets

Project Identity

Project Title

The Last Mile Challenge: Uncovering Delivery Delays, Bottlenecks, and Operational Risks Through Data Analytics

Project Overview

The Last Mile Challenge is a logistics analytics project developed to investigate the operational factors influencing delivery performance within a shipment network. The project demonstrates how structured analytical techniques can transform operational data into meaningful business intelligence that supports decision-making.

Rather than concentrating on a single metric, the analysis evaluates delivery performance from multiple operational perspectives, including courier efficiency, regional bottlenecks, seasonal trends, product category behavior, and shipment characteristics. This multi-dimensional approach provides a broader understanding of the factors contributing to delivery delays.

The project combines SQL-based business analysis with Python visualizations to present findings in a format suitable for both technical audiences and business stakeholders. Every stage of the workflow was designed to reflect practical analytical thinking commonly applied in business analytics projects.

Executive Summary

Last-mile delivery represents one of the most operationally demanding stages of the supply chain. Delivery delays not only affect customer experience but also influence operational efficiency, resource utilization, and service-level performance. Identifying the underlying drivers behind these delays is therefore essential for organizations seeking to improve logistics operations.

This project analyzes approximately 9,951 shipment records to examine delivery performance across multiple operational dimensions. The analysis investigates courier performance, city-level bottlenecks, zone performance, monthly delivery trends, product category behavior, and shipment weight characteristics to determine where operational risks are concentrated.

The analytical workflow began with data validation and cleaning to ensure data quality before business analysis. SQL was then used to answer operational business questions through structured queries, while Python was used to convert analytical outputs into executive-style visualizations that communicate findings more effectively.

Instead of treating each analysis independently, the project connects insights across different operational dimensions to identify recurring patterns and prioritize areas with the greatest business impact. This integrated approach supports evidence-based recommendations for improving delivery performance and operational efficiency.

The final project deliverables include SQL scripts, cleaned datasets, analytical reports, Python visualization scripts, executive reports, business recommendations, and portfolio-ready documentation that demonstrates both technical implementation and business interpretation.

Business Background

Logistics organizations generate large volumes of shipment data throughout their daily operations. Every shipment records information about delivery timelines, courier partners, operational regions, product categories, shipment characteristics, and delivery outcomes. While this information provides visibility into operational activities, extracting meaningful business insights requires systematic analysis.

Delivery delays often originate from multiple operational factors rather than a single cause. A city experiencing high delivery delays may simply process a small number of shipments, while another city handling thousands of orders may create a far greater operational impact despite having only moderately higher delays. Similarly, certain product categories may consistently contribute more to service-level failures due to shipment volume rather than poor delivery performance alone.

These operational relationships cannot be identified through isolated metrics or summary statistics. They require analysis across multiple business dimensions to distinguish statistically interesting observations from operationally significant findings.

This project was undertaken to investigate those relationships using historical shipment data. By combining SQL-based analytical techniques with business interpretation, the project aims to identify measurable operational bottlenecks, understand their business impact, and recommend areas where improvement efforts should be prioritized.

Rather than simply describing historical performance, the analysis seeks to answer a practical business question:

If management had to prioritize only a few operational improvements, where should those efforts be focused to achieve the greatest overall impact on delivery performance?

Problem Statement

Delivery performance is one of the most critical indicators of operational efficiency within a logistics network. Even relatively small increases in delivery delays can lead to missed service-level commitments, increased operational costs, reduced customer satisfaction, and inefficient resource utilization.

Although organizations continuously monitor shipment activity, identifying the underlying operational causes of delivery delays is significantly more challenging than simply measuring delay frequency. Delays may originate from multiple factors, including courier performance, geographical regions, shipment characteristics, seasonal demand patterns, or product-specific operational challenges.

The primary objective of this project was to investigate these operational dimensions using historical shipment data and identify which factors contributed most significantly to delivery performance.

Instead of assuming where operational issues existed, the analysis was driven through business questions designed to validate observations using data. The project focused on understanding not only where delays occurred, but also whether those delays represented meaningful business concerns when considered alongside shipment volume and operational impact.

The analysis ultimately aimed to answer questions such as:

- Are delivery delays concentrated within specific courier partners?
- Which cities and operational zones represent the greatest logistical bottlenecks?
- Do seasonal patterns influence delivery performance?
- Which product categories contribute most significantly to delivery delays?
- Does shipment weight affect delivery efficiency?
- Which operational areas should management prioritize for improvement?

By answering these questions through structured analysis, the project seeks to provide practical business recommendations rather than descriptive reporting alone.

Dataset Documentation

The analysis was performed using a shipment delivery dataset consisting of approximately **9,951 shipment records** representing logistics operations across multiple cities, operational zones, courier partners, product categories, and shipment characteristics.

Each record represents a single shipment and contains operational information required to evaluate delivery performance throughout the logistics process.

The dataset includes business attributes covering:

Shipment Information

- Order ID

- Order Date
- Promised Delivery Date
- Delivered Date
- Delivery Status
- Delay Days

Operational Attributes

- Courier Partner
- City
- Operational Zone

Product Information

- Product Category
- Shipment Weight

These attributes enable the analysis of delivery performance from multiple operational perspectives rather than relying on a single performance metric.

The dataset provides sufficient information to evaluate delivery delays geographically, operationally, seasonally, and categorically, making it suitable for exploratory business analysis and executive reporting.

Data Quality Assessment

Before beginning business analysis, the dataset underwent a structured quality assessment to ensure that analytical results would be based on reliable information.

The quality assessment focused on identifying issues that could influence business conclusions or produce misleading analytical results.

The following validation activities were performed:

Record Validation

The imported dataset was validated after loading into SQL to ensure that the expected number of records had been imported successfully without data loss.

The imported record count matched the source dataset.

Duplicate Validation

Duplicate shipment records were evaluated using the Order ID, which represents the business identifier for individual shipments.

No duplicate Order IDs were identified, confirming that each shipment record represented a unique business transaction.

Missing Value Assessment

Critical business columns were inspected for missing or blank values, including courier names, operational locations, zones, product categories, and shipment dates.

During assessment, a small number of blank categorical values and invalid placeholder dates were identified and corrected prior to analysis.

Date Validation

The shipment timeline forms the foundation of delivery performance analysis. Therefore, the three operational date fields—

- Order Date
- Promised Date
- Delivered Date

—were validated to ensure consistency.

During validation, placeholder values represented as **0000-00-00** were identified within the dataset. These values did not represent valid business dates and therefore required correction before any time-based analysis could be performed.

The invalid date values were converted to SQL NULL values to accurately represent missing information while preventing incorrect calculations during monthly trend analysis.

Data Consistency Checks

Operational dimensions such as courier names, cities, zones, delivery status, and product categories were reviewed to identify inconsistencies that could fragment analytical results.

This validation ensured that identical business entities were represented consistently throughout the dataset before aggregation and reporting.

Data Cleaning Strategy

Data cleaning was performed prior to business analysis to improve data reliability and ensure that SQL queries produced accurate business insights.

Unlike spreadsheet-based cleaning alone, the final cleaning process was implemented within SQL so that the dataset remained analysis-ready throughout the project workflow.

The cleaning strategy focused on preserving valid business records while correcting data quality issues that could affect reporting accuracy.

The primary cleaning activities included:

Handling Invalid Dates

Placeholder dates stored as **0000-00-00** were converted to SQL NULL values.

This approach ensured that missing shipment dates were represented correctly without introducing incorrect time-based calculations during monthly analysis.

Handling Blank Categorical Values

Blank values identified within categorical columns were replaced with NULL values where appropriate.

Representing missing categorical information using NULL provides a more accurate interpretation during SQL analysis than treating blanks as valid categories.

Duplicate Verification

Shipment records were examined for duplicate Order IDs.

Since no duplicate business records were identified, no deletion operations were required.

This verification step ensured that shipment counts, averages, and business KPIs were not artificially inflated.

Data Validation After Cleaning

Following the cleaning process, validation queries were executed to confirm that:

- Invalid dates had been successfully converted.
- Blank values had been corrected.
- Shipment counts remained unchanged.
- No additional inconsistencies had been introduced during cleaning.

Only after completing these validation steps was the dataset considered suitable for business analysis.

Technical Implementation

The project was implemented using a structured analytical workflow designed to mirror how operational data is investigated in a business environment. Rather than performing ad hoc analysis, each stage was completed sequentially to ensure that business conclusions were supported by reliable and validated data.

Microsoft Excel was initially used to inspect the raw dataset and understand its structure before importing the data into MySQL Workbench. SQL then became the primary analytical tool for validating data quality, cleaning invalid records, and answering business questions through structured queries.

Following SQL analysis, the query outputs were exported as CSV files and used as inputs for Python. Python was employed to transform analytical results into executive-style visualizations using Pandas,

Matplotlib, and Seaborn. These visualizations complemented the SQL findings by presenting operational trends in a format suitable for business stakeholders.

The final stage involved translating analytical findings into executive reports and business recommendations. Rather than reporting numerical outputs alone, each analysis was interpreted from an operational perspective to identify business risks, prioritize improvement areas, and communicate actionable insights.

This workflow demonstrates a complete business analysis process in which raw operational data is progressively transformed into decision-support information.

Project Execution Workflow

The project followed a structured analytical workflow consisting of six major phases.

Phase 1 – Data Understanding

The dataset was initially reviewed to understand its business context, available attributes, and potential analytical opportunities. Each column was examined to determine how it could contribute to answering operational business questions.

Understanding the dataset before writing SQL queries ensured that the analysis remained business-driven rather than query-driven.

Phase 2 – Data Validation and Cleaning

Before any analytical queries were written, the dataset underwent systematic validation.

This phase focused on:

- Record verification
- Duplicate validation
- Missing value assessment
- Invalid date correction
- Data consistency checks

Performing these validation activities first ensured that subsequent analytical results would accurately represent business operations.

Phase 3 – SQL Business Analysis

Once the dataset had been validated, SQL was used to investigate operational performance across multiple business dimensions.

Rather than writing queries without purpose, every SQL query was developed to answer a specific business question.

The analysis focused on five primary operational areas:

- Courier Performance
- City and Zone Performance
- Monthly Operational Trends
- Product Category Performance
- Shipment Weight Analysis

Each analytical section concluded with business interpretation and recommendations instead of stopping at numerical results.

Phase 4 – Python Visualization

The SQL result sets were exported as CSV files and used as input for Python.

Instead of querying the database directly from Python, exported analytical datasets were visualized using Pandas together with Matplotlib and Seaborn.

Different visualization types were selected based on the nature of each business question.

Examples include:

- Horizontal bar charts for ranked operational comparisons.
- Scatter plots for identifying relationships between shipment volume and delivery delays.
- Line charts for analysing monthly operational trends.
- Column charts for comparing category-level and shipment-weight performance.

Selecting visualization types according to the business question improved the clarity of analytical communication and reduced unnecessary visual complexity.

Phase 5 – Business Reporting

Individual analytical findings were documented immediately after completing each analysis.

Rather than waiting until the end of the project, business observations, operational interpretations, and recommendations were recorded while the analysis was still fresh.

This approach improved documentation quality and ensured that analytical reasoning was preserved throughout the project.

Phase 6 – Executive Documentation

The final phase consolidated all technical outputs, analytical findings, business recommendations, reports, visualizations, and project assets into a portfolio-ready repository.

The objective of this phase was not simply to archive project files but to create reusable documentation that supports portfolio presentation, technical interviews, resume preparation, and future project enhancements.

Business Questions

Instead of exploring the dataset randomly, the project was structured around operational business questions that reflect the type of investigations commonly performed by business analysts and logistics operations teams.

Each question was designed to evaluate a specific aspect of delivery performance and to identify whether observed patterns represented meaningful operational concerns.

The business questions investigated during the project included:

Courier Performance Analysis

Objective

To evaluate the delivery performance of individual courier partners and determine whether operational delays were concentrated within specific providers.

Business Questions

- Which courier partners handle the highest shipment volumes?
- What percentage of deliveries fall into each delivery status for every courier?
- Which courier records the highest proportion of critical delivery delays?

City and Zone Analysis

Objective

To identify geographical areas where operational bottlenecks have the greatest business impact.

Business Questions

- Which cities experience the highest average delivery delays?
- Which cities contribute the highest number of critical delays?
- Which operational zones demonstrate the poorest delivery performance?
- Should management prioritize cities with the highest delays or cities handling the largest shipment volumes?

Monthly Trend Analysis

Objective

To investigate seasonal operational behaviour throughout the year.

Business Questions

- How does shipment volume vary across months?
- Which months experience the highest average delivery delays?

- Which months contribute the greatest number of critical delivery failures?
-

Category Analysis

Objective

To evaluate whether product categories influence delivery performance.

Business Questions

- Which product categories record the highest average delivery delays?
 - Which categories contribute the highest number of critical delivery delays?
 - Which categories account for the largest proportion of shipments?
-

Weight Analysis

Objective

To determine whether shipment weight influences delivery performance.

Business Questions

- Do heavier shipments experience longer delivery delays?
- Which shipment weight category contributes the largest number of critical delays?
- Which shipment weight category dominates overall shipment volume?

These business questions formed the foundation of the SQL analysis and ensured that every query addressed a practical operational problem rather than simply exploring the dataset descriptively.

SQL Business Analysis

The analytical phase of the project was conducted using SQL, with each query designed to answer a specific business question rather than simply exploring the dataset. Instead of focusing on technical complexity, the emphasis was placed on solving operational problems that are relevant in a logistics environment.

The analysis was performed sequentially, with insights from one investigation guiding the next. This approach helped build a complete understanding of delivery performance instead of producing isolated observations.

Each analytical section follows the same methodology:

- Define the business objective.
- Develop SQL queries to answer the business question.
- Validate the analytical results.
- Interpret the findings from an operational perspective.
- Translate findings into business recommendations.

This methodology ensured that SQL served as a decision-support tool rather than merely a querying language.

Analysis 1 – Courier Performance Analysis

Business Objective

Courier partners represent one of the most important operational components within a logistics network. Differences in delivery performance between courier partners can directly influence customer satisfaction, service-level compliance, and operational efficiency.

The objective of this analysis was to determine whether delivery delays were concentrated within specific courier partners or represented a broader operational issue affecting the logistics network.

Analytical Approach

The analysis was performed in multiple stages.

Initially, shipment volumes handled by each courier partner were examined to understand operational workload distribution.

The next stage evaluated the percentage distribution of delivery statuses—including On-Time, Slight Delay, Moderate Delay, Critical Delay, and Early Delivery—for each courier.

Finally, critical delay percentages were compared across courier partners to identify whether any provider demonstrated significantly poorer performance.

Using percentages instead of raw counts allowed courier performance to be compared fairly despite differences in shipment volume.

Key Findings

The analysis revealed that Slight Delay represented the dominant delivery status across every courier partner, indicating that delivery delays were not isolated to a single provider.

Critical delivery percentages remained relatively similar across all courier partners, with only minor variation observed.

Although individual couriers demonstrated slightly higher critical delay percentages than others, the overall distribution suggested that delivery performance issues were systemic rather than courier-specific.

Business Interpretation

If only raw shipment counts had been considered, high-volume couriers would naturally appear to perform worse.

By normalizing delivery status through percentage analysis, it became evident that operational performance remained broadly consistent across courier partners.

This finding shifted the analytical focus away from blaming individual courier providers and toward investigating other operational dimensions that might contribute more significantly to delivery delays.

Business Recommendation

Rather than replacing courier partners based solely on shipment counts, management should investigate operational processes that influence all courier providers equally.

Improvements in dispatch planning, route optimization, warehouse processing, or regional logistics are likely to produce greater performance gains than focusing exclusively on courier selection.

Analysis 2 – City and Zone Bottleneck Analysis

Business Objective

Geographical analysis was conducted to determine whether delivery delays were concentrated within specific cities or operational zones.

The objective extended beyond identifying locations with the highest delays. It also considered shipment volume to distinguish statistically high delays from locations that represented meaningful business priorities.

Analytical Approach

Average delivery delays were calculated for each city to evaluate operational performance.

Critical delivery counts were analysed separately to identify locations where severe service-level failures occurred most frequently.

Shipment volume was then incorporated into the analysis to determine whether highly delayed cities also represented major operational workloads.

Zone-level analysis provided a broader regional perspective by aggregating cities into operational zones.

Key Findings

The analysis identified Kochi as the city with the highest average delivery delay.

However, further investigation showed that Kochi processed comparatively fewer shipments than cities such as Bangalore, Delhi, and Hyderabad.

Bangalore handled the highest shipment volume within the dataset, while Delhi combined relatively high delivery delays with a substantially larger operational workload.

At the regional level, North and West zones demonstrated comparatively higher delivery delays, whereas the Central zone processed the highest shipment volume.

Business Interpretation

One of the most important analytical lessons from this investigation was that average delay alone does not determine business priority.

Cities handling only a small number of shipments may exhibit high average delays without creating significant operational risk.

Conversely, cities processing thousands of deliveries can generate much greater business impact even when average delays are only moderately elevated.

This distinction illustrates the importance of combining performance metrics with operational volume before making business decisions.

Business Recommendation

Operational improvement initiatives should prioritize high-volume cities that also demonstrate elevated delivery delays, particularly Bangalore and Delhi.

Regional performance monitoring should focus on North and West zones to identify recurring operational bottlenecks and support proactive resource planning.

Analysis 3 – Monthly Trend Analysis

Business Objective

Delivery performance may vary throughout the year due to changing operational workloads, seasonal demand, or external business conditions.

The objective of monthly trend analysis was to identify periods during which logistics operations experience increased delivery pressure.

Analytical Approach

Monthly shipment volumes were calculated to understand workload distribution across the calendar year.

Average delivery delay was analysed separately to identify periods of reduced operational efficiency.

Critical delivery counts were then evaluated to determine whether specific months experienced elevated service-level failures.

By combining these three perspectives, the analysis provided a more comprehensive understanding of seasonal operational behaviour than any individual metric could provide independently.

Key Findings

Monthly trend analysis demonstrated that delivery performance varied throughout the year rather than remaining constant.

March recorded the highest average delivery delay among all months analysed, indicating a temporary decline in operational efficiency during that period.

August processed the largest shipment volume, making it the busiest operational month within the dataset.

October recorded the highest number of critical delivery delays, suggesting that periods of increased operational workload may contribute to higher service-level failures.

Although monthly differences in average delay were relatively small, analysing shipment volume alongside delay metrics revealed periods where operational pressure was significantly greater.

Business Interpretation

Evaluating only average delivery delay would have highlighted March as the poorest-performing month.

However, combining average delay with shipment volume and critical delivery counts provided a more meaningful operational perspective.

For example, August handled the highest shipment volume while maintaining delay levels comparable to several other months. In contrast, October experienced the greatest number of critical delivery failures despite not processing the largest shipment volume.

This demonstrates that operational planning should consider multiple performance indicators simultaneously rather than relying on a single KPI.

Business Recommendation

Operational planning teams should closely monitor logistics performance during high-volume months, particularly between June and October.

Resource allocation, workforce planning, and transportation capacity should be reviewed before anticipated peak periods to minimise service-level failures and maintain delivery performance.

Analysis 4 – Product Category Analysis

Business Objective

Different product categories often exhibit varying delivery characteristics due to differences in packaging requirements, handling procedures, storage conditions, and shipment demand.

The objective of this analysis was to determine whether delivery performance differed across product categories and to identify categories that represented the greatest operational risk.

Analytical Approach

Three complementary analyses were performed.

First, the average delivery delay was calculated for each product category.

Second, the number of critical delivery delays was evaluated to identify categories contributing most significantly to severe operational failures.

Finally, shipment volume distribution across categories was analysed to determine the relative operational importance of each category.

Evaluating these measures together ensured that business recommendations considered both operational performance and business impact.

Key Findings

Books recorded the highest average delivery delay among all product categories.

Electronics contributed the highest shipment volume within the dataset and also recorded the largest number of critical delivery delays.

Clothing and Grocery represented additional high-volume categories, contributing substantially to overall operational workload.

The analysis demonstrated that categories with the greatest business impact were not necessarily those with the highest average delay but those combining high shipment volume with elevated service-level failures.

Business Interpretation

The Electronics category emerged as the most operationally significant product group.

Although Books experienced the highest average delay, Electronics generated considerably greater business impact because of its large shipment volume and higher number of critical delivery failures.

This distinction reinforces an important analytical principle observed throughout the project: operational priorities should be determined by business impact rather than isolated performance metrics.

Business Recommendation

Management should prioritise process improvements within the Electronics category due to its contribution to both shipment volume and critical delivery failures.

Additional monitoring of high-volume categories can significantly improve overall delivery performance and reduce customer-facing service disruptions.

Analysis 5 – Shipment Weight Analysis

Business Objective

Shipment weight influences transportation planning, loading operations, vehicle utilisation, and handling complexity.

The purpose of this analysis was to determine whether shipment weight affected delivery performance and whether heavier shipments required additional operational attention.

Analytical Approach

Shipments were grouped into three operational categories:

- Light (<2 kg)
- Medium (2–5 kg)
- Heavy (>5 kg)

For each weight group, the following metrics were analysed:

- Average delivery delay
- Critical delivery count
- Share of total shipment volume

This approach allowed delivery performance to be evaluated alongside operational workload.

Key Findings

Heavy shipments recorded the highest average delivery delay.

However, heavy shipments represented only a very small proportion of the overall shipment volume.

Light-weight shipments accounted for more than four-fifths of all deliveries and consequently generated the highest number of critical delivery failures.

The findings indicate that shipment volume had a greater influence on operational impact than shipment weight alone.

Business Interpretation

At first glance, heavy shipments appeared to present the greatest operational challenge due to their higher average delay.

Further analysis showed that their limited shipment volume significantly reduced their overall business impact.

Light-weight shipments, despite exhibiting lower average delays, represented the majority of operational workload and therefore contributed more substantially to service-level failures.

This analysis once again demonstrated the importance of interpreting analytical results within their business context rather than relying solely on individual performance metrics.

Business Recommendation

Operational improvement initiatives should prioritise high-volume shipment categories before focusing on relatively infrequent heavy shipments.

Monitoring operational efficiency within light-weight shipment processing can produce greater improvements in overall delivery performance than concentrating exclusively on heavy-load logistics.

Cross-Analysis Insights

Each analytical section investigated delivery performance from a different operational perspective. While the individual analyses answered specific business questions, the greatest value emerged when findings were interpreted collectively.

Several recurring patterns became evident across multiple analyses.

Operational Impact Should Be Evaluated Alongside Shipment Volume

One of the strongest observations throughout the project was that operational priorities cannot be determined using performance metrics alone.

Cities, product categories, or shipment groups with the highest average delays did not always represent the greatest business concern. Business impact increased significantly when elevated delays occurred alongside high shipment volumes.

For example, although Kochi recorded the highest average delivery delay, its comparatively lower shipment volume reduced its overall operational significance. Conversely, Bangalore and Delhi handled substantially larger shipment volumes, making improvements within these locations likely to influence a much larger proportion of deliveries.

A similar pattern was observed during shipment weight analysis. Heavy shipments recorded the highest average delay but represented only a small fraction of overall shipments. Light-weight shipments dominated operational workload and therefore contributed the greatest number of critical delivery failures.

This project reinforced the importance of combining performance indicators with operational volume before establishing business priorities.

Delivery Delays Were Distributed Across Multiple Operational Dimensions

The analysis found no evidence that delivery delays originated from a single operational source.

Courier performance remained relatively consistent across providers, suggesting that delays were not isolated to one logistics partner.

Instead, operational pressure appeared to be distributed across multiple dimensions, including geography, seasonal demand, shipment characteristics, and product categories.

This finding indicates that sustainable operational improvement requires coordinated process optimization rather than isolated corrective actions.

Multiple Metrics Produce Better Business Decisions

Throughout the project, business decisions were intentionally based on multiple complementary metrics rather than a single KPI.

For example:

- Average delivery delay measured operational efficiency.
- Shipment volume measured operational importance.

- Critical delivery counts measured service-level risk.

Evaluating these measures together produced more balanced business conclusions than analysing each metric independently.

This analytical approach reduced the likelihood of prioritising statistically interesting observations that offered limited operational benefit.

Strategic Business Recommendations

The recommendations presented below are derived from the combined findings of all analyses rather than any single SQL query.

Recommendation 1

Prioritize Improvements Within High-Volume Operational Locations

Operational improvements should first target cities handling the largest shipment volumes while also demonstrating elevated delivery delays.

Reducing delays within these locations is expected to produce greater overall business impact than focusing on smaller operational regions.

Recommendation 2

Strengthen Seasonal Operational Planning

Monthly analysis indicates that logistics workload varies throughout the year.

Historical shipment patterns should be incorporated into operational planning to improve workforce allocation, transportation scheduling, and resource utilization during periods of increased demand.

Proactive planning before peak operational months can reduce service-level failures while improving delivery consistency.

Recommendation 3

Increase Monitoring of High-Impact Product Categories

Product categories contributing both substantial shipment volume and elevated delivery failures should receive additional operational attention.

Regular category-level performance reviews can help identify recurring operational issues before they affect overall delivery performance.

Recommendation 4

Continue Monitoring Regional Performance

Regional performance should be reviewed periodically rather than only after operational issues become visible.

Monitoring delivery performance at both city and zone levels enables earlier identification of emerging bottlenecks and supports proactive operational intervention.

Recommendation 5

Adopt Multi-KPI Performance Monitoring

Operational decisions should not rely on a single performance measure.

Performance reviews should incorporate multiple indicators, including shipment volume, average delivery delay, and critical delivery counts, to provide a more complete understanding of logistics performance.

Using multiple KPIs together improves decision quality and supports more effective prioritization of operational improvement initiatives.

Key Performance Indicators (KPIs)

The project evaluated delivery performance using several operational KPIs. Each KPI measured a different aspect of logistics performance and contributed to the overall analytical framework.

KPI	Business Purpose
Total Shipments	Measures operational workload handled by the logistics network.
Average Delivery Delay	Evaluates overall delivery efficiency.
Critical Delivery Count	Measures severe service-level failures requiring operational attention.
Delivery Status Distribution	Assesses delivery outcomes across different performance categories.
Shipment Percentage	Identifies the relative operational contribution of each business segment.

These KPIs were selected because they collectively measure operational efficiency, service reliability, and business impact rather than focusing on a single performance indicator.

Business Value Delivered

This project demonstrates how operational shipment data can be transformed into practical business intelligence using structured analytical techniques.

Instead of producing descriptive reports alone, the analysis identified measurable operational bottlenecks, evaluated business impact across multiple dimensions, and translated technical findings into actionable recommendations.

The resulting insights provide a stronger foundation for operational decision-making than isolated performance metrics and illustrate how SQL and Python can be used together to support evidence-based business analysis.

Challenges Faced

Every data analytics project presents both technical and analytical challenges. Throughout this project, several practical issues were encountered that required careful evaluation before meaningful business analysis could begin.

Rather than treating these as obstacles, each challenge became an opportunity to improve the analytical process and strengthen the reliability of the final conclusions.

Data Quality Challenges

The first challenge involved ensuring that the imported dataset accurately represented the original source data.

Before beginning analysis, record counts were verified between the source dataset and the SQL database to confirm that no records were lost during the import process.

During data validation, invalid placeholder dates represented as **0000-00-00** were identified within multiple date columns. These values could not be used for chronological analysis and therefore had to be converted into SQL NULL values before monthly trend analysis could be performed.

Blank categorical values were also identified and standardized to prevent them from being treated as valid business categories during aggregation.

These validation activities ensured that subsequent analyses were performed on reliable and consistent data.

Distinguishing Business Significance from Statistical Observations

One of the most important analytical challenges involved interpreting results correctly rather than relying solely on numerical rankings.

For example, cities with the highest average delivery delays initially appeared to represent the largest operational problems.

Further investigation revealed that some of these cities processed relatively few shipments, meaning that their operational impact was considerably smaller than cities handling much larger delivery volumes.

This highlighted the importance of combining performance metrics with shipment volume before drawing business conclusions.

A similar challenge emerged during shipment weight analysis, where heavier shipments recorded longer average delays but represented only a very small proportion of the overall logistics workload.

These situations reinforced the importance of evaluating business context alongside statistical measures.

Selecting Meaningful Business Questions

Another challenge involved deciding which analytical questions would provide genuine business value.

Instead of exploring every available column, the project focused on operational questions that management could realistically use for decision-making.

This required continuously asking whether each analysis contributed to a better understanding of delivery performance or merely produced additional descriptive statistics.

Restricting the analysis to business-relevant questions improved both the clarity and practical value of the project.

Converting Technical Results into Business Insights

Producing SQL query results represented only one stage of the analytical process.

The greater challenge was translating numerical outputs into operational insights that could support business decisions.

Rather than describing SQL outputs directly, each analytical section was interpreted from a business perspective by considering operational impact, shipment volume, service-level performance, and potential improvement opportunities.

This approach ensured that technical analysis remained closely aligned with business objectives.

Lessons Learned

This project significantly strengthened both technical capability and analytical thinking.

While SQL, Python, and visualization tools were essential components of the implementation, the most valuable learning outcomes were related to structured problem solving and business interpretation.

Data Validation Must Precede Analysis

One of the strongest lessons learned was that reliable business insights depend on reliable data.

Even relatively small issues such as invalid dates or blank categorical values can influence aggregation results, trend analysis, and KPI calculations.

Systematic data validation should therefore be considered an essential stage of every analytical workflow rather than an optional preprocessing step.

Business Context Is More Important Than Individual Metrics

Throughout the project, multiple situations demonstrated that isolated performance metrics can produce misleading business conclusions.

Average delay alone was insufficient for determining operational priorities.

Shipment volume, critical delivery counts, and operational significance had to be considered together before identifying the areas requiring management attention.

This reinforced the importance of analysing business impact rather than simply identifying statistical extremes.

SQL Is a Business Decision Tool

Initially, SQL appeared to be a language for retrieving data.

As the project progressed, it became clear that SQL is equally valuable as a tool for investigating business problems, validating assumptions, and supporting operational decision-making.

The effectiveness of a SQL query depends not only on its syntax but also on the quality of the business question it is designed to answer.

Effective Visualizations Support Decision-Making

Python visualizations were created to communicate analytical findings more effectively.

An important lesson learned during this stage was that selecting an appropriate visualization depends on the business question being answered rather than personal preference.

Different chart types were intentionally selected to highlight comparisons, trends, rankings, and relationships more clearly.

This improved the readability of the analytical results and strengthened the overall business narrative.

Documentation Is Part of the Analytics Process

The project demonstrated that documentation should not be treated as a final administrative activity.

Recording business objectives, analytical reasoning, findings, and recommendations throughout the project preserved important decisions that would otherwise be forgotten over time.

Comprehensive documentation also improves knowledge transfer, project reproducibility, and portfolio quality.

Overall Project Reflection

The Last Mile Challenge represents more than an exercise in SQL querying or Python visualization.

The project demonstrates a structured analytical approach in which business problems are investigated systematically, data quality is validated before analysis, technical findings are interpreted within their operational context, and recommendations are supported by evidence rather than assumptions.

More importantly, the project strengthened the ability to think beyond query writing by focusing on why a business question is important, how it should be investigated, and how analytical results can contribute to better operational decisions.

These experiences establish a strong foundation for applying similar analytical methodologies to future business domains and more complex real-world datasets.

Project Architecture

Raw CSV Dataset

|



Data Validation (Excel + SQL)

|



SQL Data Cleaning

|



Business Analysis

|

├— Courier Analysis

├— City & Zone Analysis

├— Monthly Trends

├— Category Analysis

└— Weight Analysis

|



Export Query Results (CSV)

|



Python Visualizations

|



Executive Reports



GitHub Portfolio

STAR Interview Version

The following STAR response summarizes the project in a structured format commonly used during behavioural and technical interviews.

Situation

The logistics industry generates large amounts of operational data every day, but identifying the actual causes of delivery delays requires more than simply measuring shipment performance. Different operational factors such as courier partners, geographical regions, product categories, shipment characteristics, and seasonal demand can all influence delivery outcomes. I wanted to build a portfolio project that demonstrated not only SQL querying skills but also the ability to investigate a real business problem using structured analytical thinking.

Task

My objective was to analyze approximately 9,951 shipment records to identify the major operational bottlenecks affecting delivery performance.

The project required me to:

- Validate and clean the dataset.
 - Design meaningful business questions.
 - Perform SQL-based business analysis.
 - Visualize analytical findings using Python.
 - Translate technical outputs into business insights and recommendations suitable for management.
-

Actions

I began by understanding the dataset structure and validating the imported records to ensure data integrity.

I performed data quality assessment by checking duplicate records, blank values, invalid dates, and overall data consistency before beginning any analysis.

After preparing the dataset, I developed SQL queries to investigate delivery performance from multiple business perspectives, including courier performance, city and zone bottlenecks, seasonal delivery trends, product category performance, and shipment weight analysis.

Instead of interpreting query results independently, I compared multiple KPIs such as shipment volume, average delivery delay, and critical delivery counts to determine which operational issues represented the greatest business impact.

The SQL outputs were exported and visualized using Python with Pandas, Matplotlib, and Seaborn to create executive-style charts that improved the communication of analytical findings.

Finally, I documented every stage of the project, including technical implementation, analytical reasoning, business interpretation, recommendations, and executive insights.

Result

The project successfully identified several operational patterns affecting delivery performance.

The analysis demonstrated that operational priorities should be determined using multiple business metrics rather than isolated statistics.

High-volume operational locations, seasonal workload variations, and specific product categories emerged as higher-priority improvement areas than observations based solely on average delay.

Beyond the analytical findings, the project strengthened my practical experience in data validation, SQL-based business analysis, Python visualization, business reporting, and analytical storytelling while producing a portfolio-ready project suitable for technical interviews and professional presentation.

Resume Master Description

Developed a comprehensive logistics analytics project to investigate delivery performance using approximately 9,951 shipment records. Performed data validation and cleaning using SQL, followed by business analysis across courier performance, geographical bottlenecks, seasonal delivery trends, product category performance, and shipment weight characteristics. Exported analytical results for visualization using Python (Pandas, Matplotlib, and Seaborn) and created executive-style reports translating technical findings into actionable business recommendations. Demonstrated proficiency in SQL querying, data cleaning, KPI analysis, business interpretation, data visualization, analytical documentation, and executive reporting while applying structured problem-solving techniques to a real-world logistics case study.

ATS Keywords

SQL

MySQL

MySQL Workbench

Python

Pandas

Matplotlib

Seaborn

Microsoft Excel

Data Cleaning
Data Validation
Business Analytics
Logistics Analytics
Operational Analytics
Data Visualization
Exploratory Data Analysis (EDA)
KPI Analysis
Business Intelligence
Root Cause Analysis
Trend Analysis
Business Reporting
Executive Reporting
Analytical Thinking
Problem Solving
Documentation
Git
GitHub
Portfolio Project

Skills Demonstrated

Technical Skills

- SQL Query Development
- Data Cleaning
- Data Validation
- Python Programming
- Pandas
- Matplotlib
- Seaborn
- Microsoft Excel
- Git & GitHub

Analytical Skills

- Business Problem Solving
 - Exploratory Data Analysis
 - KPI Design and Interpretation
 - Trend Analysis
 - Root Cause Investigation
 - Business Recommendation Development
 - Multi-dimensional Performance Analysis
-

Professional Skills

- Executive Reporting
 - Technical Documentation
 - Business Communication
 - Analytical Storytelling
 - Structured Problem Solving
 - Attention to Data Quality
 - Portfolio Development
-

Future Enhancements

Several opportunities exist to extend this project beyond its current scope.

Potential future improvements include developing an interactive Power BI dashboard to enable dynamic exploration of delivery performance metrics, integrating predictive models for delivery delay forecasting, automating SQL-to-Python reporting workflows, and incorporating additional operational variables such as weather conditions, traffic information, warehouse processing time, or customer feedback.

These enhancements would allow the project to evolve from descriptive business analytics toward predictive and decision-support analytics.

Portfolio Assets

The repository contains the following project deliverables:

- Original and cleaned datasets
- SQL scripts for data import, validation, cleaning, and business analysis

- Exported analytical result sets
- Python visualization script (logistics_dashboard.py)
- Executive-style analytical charts
- Individual analysis reports
- Executive Insights Report
- README documentation
- Project Master Documentation

Collectively, these assets demonstrate the complete analytical workflow from raw operational data to business recommendations and portfolio-ready documentation.